

Yiqiao Jin

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EDUCATION

Washington University in St. Louis, Department of Statistics & Data Science Ph.D. in Statistics Advisor: Dr. Mengxin (Maxine) Yu & Dr. Xiaofeng Shao	St. Louis, MO Aug 2024 – May 2029 (Expected)
Fudan University, School of Data Science Master of Science in Applied Statistics Advisor: Dr. Bo Fu	Shanghai, China Aug 2022 – Jun 2024
Shanghai Normal University, Department of Mathematics Bachelor of Science in Statistics Advisor: Dr. Xiaoqing Pan	Shanghai, China Aug 2018 – Jun 2022

HONORS & AWARDS

Outstanding Qualifiers Performance, WashU	Apr 2026
Graduate Service and Leadership Award, Honorable Mention, WashU	Apr 2026
ENAR Distinguished Student Paper Award, International Biometric Society (ENAR)	Dec 2025

RESEARCH INTERESTS

Ranking Inference; Synthetic Data Augmentation; High-dimensional Statistics; Causal Inference; Electronic Health Records

PUBLICATIONS & MANUSCRIPTS

Jin, Y., Jeon, E., Ghosh, S. K., Devireddy, K., Garg, M., Liu, X., Geda, Y. E., St. Sauver, J., Vassilaki, M., Petersen, R. C., Sohn, S. (2026+) Estimation of Dementia Risk Under Persistent and Intermittent Neuropsychiatric Symptom Trajectories Across Age Subgroups. *Submitted*.

Devireddy, K., Garg, M., **Jin, Y.**, Ghosh, S. K., Liu, X., Jeon, E., Geda, Y. E., St. Sauver, J., Vassilaki, M., Kappor, E., Petersen, R. C., Sohn, S. (2026+) An Interpretable Continuous-Time Multistate Model for Neuropsychiatric Symptoms and Cognitive State Transitions. *Manuscript in preparation*.

Ghosh, S. K., Liu, X., Devireddy, K., **Jin, Y.**, Garg, M., Jeon, E., Geda, Y. E., St. Sauver, J., Vassilaki, M., Petersen, R. C., Sohn, S. (2026+) A Reinforcement Learning-Based Multi-Agent Framework for Personalized Dementia Risk Alerting Following Neuropsychiatric Symptom Onset. *Submitted*.

Yang, Z., **Jin, Y.**, Yu, M. (2026+) Cost-Aware Design for Target-Population Ranking from Partially Labeled Comparisons. *Manuscript in preparation*.

Jin, Y., Yu, M. (2026+) Maximum Likelihood Estimation for Entity Ranking under Iterative Synthetic Data Augmentation. *Manuscript in preparation*.

Xin, R., **Jin, Y.**, Lin, N., Zhang, L. (2026+) Federated R-learner for Estimating Conditional Average Treatment Effects across Real-World Clinical Datasets. *Manuscript in preparation*.

Xin, R., **Jin, Y.**, Lin, N., Zhang, L. (2026) Federated R-Learner for Estimating Heterogeneous Treatment Effects for Personalized Treatments. *Oral Presentation at AMIA 2026 Amplify Informatics Conference*

Jin, Y., Lin, N., Zhang, L. (2025) Federated R-Learner for Estimating Conditional Average Treatment Effect across Heterogeneous Datasets. *Winner of ENAR 2026 Distinguished Student Paper Awards*

TALKS & PRESENTATIONS

Maximum Likelihood Estimation for Entity Ranking under Iterative Synthetic Data Augmentation <i>Joint Statistical Meetings (JSM 2026), Session: Uncertainty Quantification in Generative AI</i>	Aug 2026 Boston, MA
Federated R-Learner for Estimating Conditional Average Treatment Effect across Heterogeneous Datasets <i>ENAR 2026 Spring Meeting, Session: Causal Inference for High-Dimensional Data</i>	Mar 2026 Indianapolis, IN
Maximum Likelihood Estimation for Entity Ranking under Iterative Synthetic Data Augmentation <i>Graduate Student Seminar, WashU</i>	Mar 2026 St. Louis, MO
Implicit Stochastic Gradient Descent <i>Stochastic Approximation Reading Group, WashU</i>	Apr 2025 St. Louis, MO
Introduction to Simulation-based Inference <i>Uncertainty Quantification and Synthetic Data Reading Group, WashU</i>	Feb 2025 St. Louis, MO

RESEARCH EXPERIENCE

Mayo Clinic, Department of Artificial Intelligence & Informatics <i>Data Science AI&I Intern (PhD Level), Mentors: Dr. Eunji Jeon & Dr. Sunghwan Sohn</i> <i>Project: Dementia Risk under Persistent and Intermittent Neuropsychiatric Symptom Trajectories</i>	Rochester, MN May 2026 – July 2026
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- Led a longitudinal causal analysis of dementia risk under persistent vs. intermittent neuropsychiatric symptom (NPS) trajectories, following 2,762 dementia-free participants over five visits in the Mayo Clinic Study of Aging.
- Designed clinically interpretable trajectory estimands (consecutive, intermittent, sustained) for four NPS syndromes (psychosis, hyperactivity, affect, apathy), enabling direct comparison holding the number of syndrome-positive visits fixed.
- Estimated population-level and age-specific risk ratios via a multiplicative structural nested mean model with doubly robust g-estimation.
- Showed that apathy and psychosis carry the strongest dementia-risk signals—consecutive occurrence of symptoms stronger than intermittent—concentrated in ages 60–84.

WashU, Arts & Sciences, Department of Statistics & Data Science <i>Graduate Research Assistant, Advisor: Dr. Mengxin (Marine) Yu</i> <i>Project 1: Cost-Aware Design for Target-Population Ranking from Partially Labeled Comparisons (Mar 2026 – Present)</i>	St. Louis, MO June 2025 – Present
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- Formulated leaderboard estimation for a target population from mixed-size top-choice comparisons as a target-protocol weighted Plackett–Luce projection, keeping the estimand well defined under model misspecification and separating it from the source sampling design.
- Derived the efficient influence function and a cross-fitted one-step estimator under transportability, missing-at-random labeling, overlap, and local connectivity conditions.
- Reparameterized the semiparametric efficiency bound into a convex, cost-aware design problem yielding square-root allocation rules over comparison actions and expensive label acquisition.

Project 2: Maximum Likelihood Estimation for Entity Ranking under Iterative Synthetic Data Augmentation (June 2025 – Feb 2026)

- First-authored a theory manuscript on ranking inference under the Bradley–Terry–Luce (BTL) model when training data are iteratively augmented with synthetic comparisons.
- Established finite-sample optimal ℓ_2 and ℓ_∞ statistical rates, exact top- K recovery and asymptotic normality of the iterative MLE over sparse comparison graphs, enabling uncertainty quantification for retrained ranking systems.
- Developed a novel coupled induction argument with leave-one-out constructions and quadratic proxies that disentangles temporal dependence across retraining iterations, proving that model collapse is avoided whenever the number of iterations grows at most polynomially in the number of items.
- Validated the theory through large-scale simulations and an application to the Arena Human Preference 140k dataset.

WashU Institute for Informatics, Data Science and Biostatistics <i>Graduate Research Assistant, Advisors: Dr. Nan Lin & Dr. Linying Zhang</i> <i>Project: Federated R-Learner for Estimating CATE across Heterogeneous Datasets</i>	St. Louis, MO May 2025 – Oct 2025
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- Developed Fed-R, a privacy-preserving, communication-efficient federated R-learner combining local cross-fitting with Neyman-orthogonal residualization and FedProx-style proximal aggregation, achieving $\sqrt{N}$ -consistency and asymptotic normality without sharing patient-level data.
- Uncovered a performance-reversal phenomenon: federated estimation surpasses centralized pooling once sites violate the overlap assumption, cutting global RMSE by 50–70% and improving local estimation accuracy by 2–4 $\times$, with most gains realized within 2–5 communication rounds.

- Benchmarked Fed-R against oracle and centralized R-learners across six synthetic scenarios and the semi-synthetic IHDP benchmark.
- Cleaned and analyzed a multi-site EHR cohort (90K+ patients, 60K+ covariates) comparing two antihypertensive drugs on time-to-event outcomes.

TEACHING EXPERIENCE

Assistant Instructor

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| • SDS 2020 – Elementary Probability and Statistics | Fall 2026 |
| • SDS 2020 – Elementary Probability and Statistics | Spring 2026 |

Teaching Assistant

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| • SDS 5210 – Statistical Computation | Fall 2025 |
| • SDS 5531 – Advanced Statistical Computation | Fall 2025 |